

Disaster Recovery / Infrastructure Resilience (Evidence)

Community Health Media · CHT Platform · Source: docs/compliance/disaster-recovery.md

This document summarizes backup, restore, and multi-region standby practices for the CHT Platform AWS environment.

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# Disaster Recovery Plan

Recovery procedures for the CHT Platform when infrastructure or data must be restored after failure, corruption, or region-

**Primary region:** `us-east-1`
**AWS account:** `233636046512`
**Standby:** `us-east-2` Terraform exists (`infrastructure/terraform/environments/us-east-2`) but is **not deployed by default
**Last reviewed:** `_Set date when approved_`

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## Recovery objectives

| Metric | Platform (prod) | Dev | Notes |
|-----|-----|-----|-----|
| **RPO** (max data loss) | **24 hours** | **24 hours** | Automated daily RDS snapshots; finer RPO requires increasing back
| **RTO** (max downtime) | **4 hours** | **8 hours** | Single-region, no hot standby; depends on engineer availability |
| **RTO (region failover)** | **24+ hours** | N/A | Requires deploying `us-east-2` stack and DNS cutover: not automated |

These targets are **goals** for planning and audits. Update after architecture changes (e.g. Multi-AZ RDS, active DR region

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## What is protected automatically

| Asset | Protection | Config (platform) |
|-----|-----|-----|
| **RDS PostgreSQL** | Automated daily backups + snapshots | `cht-platform-db`, retention **7 days**, window **03:00-04:00
| **RDS** | Encryption at rest (KMS), deletion protection | `deletion_protection = true` |
| **RDS** | Final snapshot on delete | `cht-platform-final-snapshot` |
| **Terraform state** | S3 versioning + encryption | `cht-platform-terraform-state` |
| **Frontend** | S3 + CloudFront; redeploy from CI | Versioned build artifacts in workflow |
| **Container images** | ECR tagged images | Tags: `v`, `platform-{sha}-{timestamp}`, `staging-latest` |
| **Secrets** | AWS Secrets Manager | Rotated via Terraform + GitHub Environment secrets |
| **Audit logs** | CloudTrail, VPC flow, ECS logs | 365-day retention (platform/staging) |

**Staging:** `cht-platform-dev-db` with **1-day** backup retention (`dev.tfvars`).

**Point-in-time recovery (PITR):** Available while `backup_retention_period > 0`. Restores to any second within the retention

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## Environment reference

### Platform (production)

| Resource | Name |
|-----|-----|
| Domain | `testapp.communityhealth.media` |
| ECS cluster | `cht-platform-cluster` |
| Backend service | `cht-platform-backend` |
| Worker service | `cht-platform-worker` |
| RDS instance | `cht-platform-db` |
| Health check | `https://testapp.communityhealth.media/health/ready` |
| ECR images | `cht-platform-backend`, `cht-platform-worker` |

### Staging

| Resource | Name |
|-----|-----|
| Domain | `devapp.communityhealth.media` |
| ECS cluster | `cht-dev-cluster` |
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| Backend service | `cht-dev-backend` |  
| Worker service | `cht-dev-worker` |  
| RDS instance   | `cht-staging-db` |  
| Health check   | `https://devapp.communityhealth.media/health/ready` |
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Scenario 1: Application failure (ECS / bad deploy)

****Symptoms:**** Health check failing, 5xx from ALB, recent deploy.

****RTO:**** ~15-30 minutes.

Option A: GitHub Actions rollback (preferred)

1. GitHub → ****Actions**** → ******